

Day 8 — Precession of the Equinoxes

Module: Movement of the Sun · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: explain why indian and western zodiac dates differ today.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Sūrya* (IAST: Sūrya; lit. “the Sun”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **The phenomenon.** Earth’s axis wobbles like a slow top, completing one full circle every ~25,800 years. Rate: ~50.3 arcseconds per year. Discovered by Hipparchus ~130 BCE.
2. **Indian computation.** Āryabhaṭīya (499 CE) implicitly includes precession in its longitude calculations. Later Indian astronomers (Bhāskara II, Nīlakaṇṭha) refine the value.
3. **Modern value.** ~50.27”/year — very close to traditional Indian estimates.
4. **Consequence for the zodiac.** The Vernal Equinox point shifts westward through the sidereal zodiac over time. ~2000 years ago, the Vernal Equinox was in Aries (which is why Aries is the “first” zodiac sign in Western astrology). Today, the Vernal Equinox is in Pisces.
5. **Modern source.** NASA / IAU references. Discussion: “*How long until the Sun is in Aquarius at the Vernal Equinox?*” (~600 years.)

Activity (15 min)

A scaled version of *Sunrise Direction Tracking* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Precession of the Equinoxes.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).